

Ain e Muhammad
Senior Software Engineer | Full-Stack | AI & 3D Systems
Albany, NY | ainemuhammad903@gmail.com | +1 (518) 542-9640
Portfolio : <https://ainemuhammad.github.io/portfolio>
LinkedIn: <https://www.linkedin.com/in/ain-m/>
Github: <https://github.com/ainemuhammad>

EXPERTISE SUMMARY

- Senior Software Engineer with extensive experience in Full-Stack development (React.js / Python), Serverless Architecture (AWS), and real-time 3D environments.
- Currently pursuing an M.S. in Computer Science (AI), specializing in Machine Learning, Computer Vision, and Generative 3D workflows.
- Hands-on experience architecting Generative AI pipelines, including leveraging state-of-the-art diffusion and 3D reconstruction models (Meshy, Tripo 3D, Hunyuan3D) alongside complex object/camera placement algorithms.
- Delivered production-ready full-stack applications with React.js, Python, and serverless AWS architectures, supporting high-concurrency workloads with minimal latency.
- Specialized in 3D Systems & Graphics Development, utilizing Unreal Engine for cinematic environments and Three.js / React Three Fiber / Blender for web-based interactive product configurators.
- Expertise in Cloud Infrastructure, specifically designing and deploying high-concurrency AWS Lambda functions, API Gateway, S3, and RDS/PostgreSQL databases.
- Proven record in Data Engineering, including dataset curation, schema design, SQL optimization, and building automated validation pipelines.
- Led and mentored junior engineers, improving team productivity and code quality through structured code reviews, CI/CD implementation, and unit testing.

EDUCATION

State University of New York (SUNY) - Albany, NY Master of Science in Computer Science
(Concentration: AI) | Expected May 2027

- **Relevant Coursework:** Machine Learning, Computer Vision, Advanced Algorithms and Data Structures.

National University of Sciences and Technology (NUST), Bachelor of Science in Software Engineering | July 2021

- **Relevant Coursework:** Artificial Intelligence, Data Warehousing and Data Mining, Data Structures and Algorithms, Database Systems, Digital Image Processing, Software Engineering, Probability and Statistics, Object-Oriented Programming, Data Structures & Algorithms, Web Engineering, Software Architecture.

WORK EXPERIENCE

3D Creative Engineering Intern – Sony Immersive Music Studio | June 2026 – Aug 2026

- **Generative 3D Pipelines:** Architected an end-to-end generative pipeline that reconstructs immersive 3D scenes from 2D images by orchestrating SOTA image segmentation (masks), diffusion-based image generation, and 2D-to-3D asset generation models.
- **Spatial & Camera Automation:** Programmed automated camera placement and layout algorithms utilizing cutting-edge spatial computing and object placement research papers, ensuring mathematically precise asset distribution within a 3D space.
- **Real-time Graphics & Environments:** Engineered interactive 3D environments and cinematic sequences within Unreal Engine, managing real-time rendering, material shaders, and complex environmental lighting.
- **AI-Assisted Scene Generation:** Designed procedural environment generation workflows leveraging Florence-2 object detection, depth-map reconstruction, Blender Python automation, and AI-generated assets to transform concept art into structured 3D scenes.

- **Virtual Production Systems:** Developed Blender-to-Unreal workflows integrating AI-generated assets, automated scene assembly, lighting analysis, camera orchestration, and cinematic rendering pipelines for immersive content creation.
- **AI Tool Development:** Built proof-of-concept Unreal Engine tooling enabling concept-art-to-3D asset generation directly within the editor, supporting local AI inference and artist-controlled asset creation workflows.
- **Scene Optimization & Rendering:** Optimized large-scale 3D environments containing hundreds of assets through instancing strategies, mesh audits, lighting refinement, and rendering pipeline improvements for real-time performance.
- **AI Agent Workflows:** Developed custom AI-assisted tooling and workflows using Claude Code, MCP integrations directly into blender, Blender automation, and Unreal Engine pipelines to accelerate environment creation, lighting design, cinematography, and content iteration.

Graduate Grader - State University of New York - Albany, NY | Jan 2026 – May 2026

- **Academic Support:** Grade assignments and exams for Discrete Structures, providing feedback on mathematical reasoning and problem-solving.
- **Course Coordination:** Support instructor with administration and provide insights into student challenges to improve curriculum delivery.

ALL3D - San Francisco, USA (Remote), May 2021 – Aug 2025

Software Engineer → Senior Software Engineer → Lead Software Engineer

Lead Software Engineer | Apr 2024 – Aug 2025

- **Client-Facing Technical Leadership:** Led delivery of 3D configurators for enterprise clients (**Walmart, Rough Country, Telescope Casual Furniture**), owning requirements, demos, and end-to-end technical execution, reducing delivery iteration cycles by ~20%.
- **Codebase Standardization & Maintainability:** Introduced coding standards, reusable patterns, and documentation practices that improved onboarding speed and reduced long-term maintenance overhead.
- **Stakeholder Communication & Technical Demos:** Delivered technical walkthroughs and demos to non-technical stakeholders, improving transparency and accelerating decision-making during project cycles.
- **Team Leadership & Execution:** Managed a cross-functional team of engineers and 3D artists, improving delivery throughput through structured task ownership and code review processes.
- **3D Rendering Optimization:** Built React Three Fiber systems leveraging GPU acceleration, GLTF compression, and LOD techniques, improving render performance and reducing load times.
- **Engineering Mentorship:** Mentored engineers, enforced architectural standards, and led code reviews, reducing production issues and improving system reliability.

Senior Software Engineer | Feb 2023 – Apr 2024

- **Serverless Architecture:** Architected AWS Lambda and API Gateway systems with RDS (PostgreSQL), improving scalability and reducing backend latency.
- **Frontend Scalability:** Developed React systems with asset streaming and lazy loading, improving initial load performance by ~35% under high concurrency.
- **3D Rendering Pipelines:** Owned backend systems for dynamic rendering (camera, lighting, scene configs), enabling flexible and reusable visualization workflows.
- **AI Experimentation:** Collaborated with AI teams on early-stage generative pipelines, improving model output quality through iterative evaluation and feedback loops.
- **System Reliability Engineering:** Implemented structured validation and error handling, reducing runtime failures and improving API stability.

Software Engineer | May 2021 – Feb 2023

- **Serverless API Development:** Built AWS Lambda APIs in Python with strong data integrity and scalability, supporting growing application workloads.
- **Frontend Engineering:** Developed React-based interfaces integrated with cloud services, improving responsiveness and user experience.

- **Cloud Asset Delivery:** Optimized 3D asset delivery via S3 and CloudFront, reducing asset load latency by ~25%.
- **Backend Automation:** Developed Python automation scripts for internal workflows, reducing manual operational overhead.
- **Testing Foundations:** Contributed to early testing infrastructure, improving code quality and deployment confidence.

Frontend Developer - Sanje Solutions | June 2020 – October 2020

- **Frontend Development:** Built responsive interfaces using React, HTML, CSS, and Bootstrap.
- **UI Implementation:** Translated design mockups into reusable, production-ready components with cross-browser compatibility.
- **Performance & UX:** Improved UI consistency and performance across client-facing applications.

Research Engineer - NUST (AI Systems & Web Systems) | June 2019 – October 2019

- **AI + Web Integration:** Developed web interfaces to visualize model outputs and enable interaction with AI systems.
- **Research Collaboration:** Worked with academic teams to translate research problems into technical implementations
- **ML Pipelines Exploration:** Explored integration of machine learning workflows with lightweight web-based systems.

PROJECTS

3D Truck Configurator

- Built an interactive 3D product configurator using React, Three.js, and React Three Fiber, enabling real-time customization of vehicle color, parts, and environment lighting via PBR materials and HDRI maps.
- Architected a cross-window postMessage API enabling seamless third-party iframe embedding, exposing commands for live color/part updates, full config load, and state reset.
- Implemented state management with Zustand and deployed to production via AWS S3 + CloudFront.

Leaf Lab

- Developed an interactive educational web app teaching kids photosynthesis through a hands-on simulation (add CO₂/water, charge sunlight, run the reaction, see glucose/oxygen output).
- Built with React, TypeScript, Vite, Tailwind CSS, and Framer Motion; set up automated GitHub Actions deployment pipeline to GitHub Pages.

Generative 3D Environment Reconstruction Platform

- Developed workflows for reconstructing immersive 3D scenes from images using AI-generated assets, depth estimation, procedural scene assembly, automated camera placement, and Unreal Engine rendering.
- Integrated Blender, Unreal Engine, Claude Code, Florence-2, and modern image-to-3D models to accelerate virtual production workflows.

Pro Studio - 3D Room Layout Editor

- Built a browser-based 3D room design editor with first-person walkthrough, orbit, and top-down navigation modes, running entirely client-side with no backend
- Implemented an interactive scene editor with transform gizmos (move/rotate/scale), collision detection, snap-to-surface placement, and full undo/redo history (100-step)
- Developed a live material editor (color, roughness, metalness) and day/night lighting system with independent light-fixture control
- Built shareable layout links by encoding full scene and lighting state into URLs, plus PNG screenshot export and local-storage persistence
- Parsed GLB 3D assets at runtime to auto-generate a furniture catalog, distinguishing fixed architecture from placeable objects by node naming convention

TECHNICAL SKILLS

- **AI & Machine Learning:** Python (Advanced), PyTorch, TensorFlow, OpenCV, Computer Vision, Deep Learning, Pandas, NumPy, Matplotlib.
- **Web & Full-Stack:** JavaScript, TypeScript, React.js, Next.js, Node.js, Material UI, SWR, HTML5, CSS3.
- **Cloud & DevOps:** AWS (Lambda, API Gateway, S3, RDS, EC2, CloudFront), Docker, Git, GitHub, CI/CD, Unit Testing (Jest/Pytest).
- **3D Visualization:** Unreal Engine , Blender, Three.js, React Three Fiber, WebGL, Blender, 3DS MAX, GLTF Compression, GPU Acceleration, Level of Detail (LOD).Unreal Engine 5, Blender Python (bpy), Gaussian Splatting, USD Pipeline, MetaHuman, Procedural Environment Generation
- **Databases:** PostgreSQL (RDS), MySQL, SQL, Schema Design, Query Optimization, Data Validation.
- **Generative AI & Spatial Models:** Generative 3D Reconstruction (Meshy, Tripo 3D, Hunyuan3D), Diffusion Models, Image Segmentation Masks, Automated Camera & Object Placement Algorithms.